

Supplementary Material

S1 Dataset, Labeling, and Data Integrity

The analysis included 20 recording sessions from 10 participants, as reported in the main manuscript. Recording duration totaled 17.40 h: 11.33 h Awake and 6.07 h Drowsy. Labels were assigned using the Karolinska Sleepiness Scale (KSS), with video observation used for cross-checking. Participant-session linkage was unavailable. Primary inference therefore used recording sessions and does not establish participant-level effects.

The data contained 1,620,038 samples. Recording session 01 was sampled at 50 Hz and the remaining sessions at approximately 25 Hz. Sampling rates were estimated from the reciprocal of the median positive timestamp interval. Integrity checks found strictly increasing timestamps, no duplicates or invalid labels, and no missing, non-numeric, or infinite signal/time values. Raw PPG, processed PPG, and labels shared the same row axis. All sessions passed integrity and analysis-eligibility checks (Table S1).

There were 69 temporal gaps exceeding 1.5 times the median sampling interval; 62 exceeded twice this interval, and none exceeded five times it. Gaps remained discontinuities without interpolation. The labels contained 86 state transitions, none across a gap. Splitting at state changes and gaps yielded 175 contiguous state segments (93 Awake, 82 Drowsy), with median duration 260.04 s; 60 were shorter than 60 s, 78 shorter than 180 s, and 94 shorter than 300 s.

Table S1: Dataset and integrity summary.

Session	Samples	Estimated fs (Hz)	Awake /Drowsy samples	Gaps	Integrity	Analysis eligible
01	117,828	50.000	77,805 /40,023	0	Pass	Yes
04	75,416	24.992	46,897 /28,519	3	Pass	Yes
05	91,211	25.003	59,235 /31,976	4	Pass	Yes
06	79,179	24.993	51,240 /27,939	5	Pass	Yes
07	79,768	24.994	59,225 /20,543	3	Pass	Yes
08	77,595	24.994	61,405 /16,190	4	Pass	Yes
09	119,272	25.001	83,631 /35,641	8	Pass	Yes
10	60,386	24.998	45,365 /15,021	3	Pass	Yes
11	61,582	24.994	33,062 /28,520	2	Pass	Yes
12	89,853	24.989	34,376 /55,477	2	Pass	Yes
13	76,473	25.000	40,272 /36,201	2	Pass	Yes
14	85,948	24.994	54,871 /31,077	5	Pass	Yes
15	69,362	24.996	54,511 /14,851	6	Pass	Yes
17	68,752	24.993	45,601 /23,151	2	Pass	Yes
18	69,937	24.997	30,634 /39,303	3	Pass	Yes
19	73,041	24.996	54,123 /18,918	4	Pass	Yes
21	69,660	25.000	41,549 /28,111	2	Pass	Yes
22	74,710	25.001	50,359 /24,351	3	Pass	Yes
23	109,452	24.997	87,703 /21,749	3	Pass	Yes
25	70,613	24.996	46,905 /23,708	5	Pass	Yes
Total	1,620,038	Mixed	1,058,769 /561,269	69	20/20	20/20

State composition is a sample count, not a duration estimate. Gaps use the threshold of 1.5 times the median timestamp interval. Analysis eligibility does not imply that every sample entered a retained window.

State-duration support in the integrity summaries used sample-associated intervals, capping gaps at the median sampling interval and assigning one median interval to the final sample. Recording-duration totals in the text follow the main manuscript.

S2 Preprocessing, Quality Control, and Retained-Window Counts

Raw PPG was multiplied by -1 and processed with a second-order Butterworth band-pass filter (0.5–8 Hz), applied forward and backward for zero phase. All subsequent QC and nonlinear analyses used processed PPG.

Artifact screening used complete, non-overlapping 5-s blocks of $\text{round}(5f_s)$ samples within each recording session. Amplitude and roughness were defined as

$$A = Q_{0.95}(x) - Q_{0.05}(x),$$

$$R = \sqrt{\text{mean}((\Delta x)^2)}.$$

Here, Q_p is the p th quantile and Δx is the first difference. Each feature $F \in \{A, R\}$ was standardized across the complete blocks of the same session:

$$z_F = 0.6745 \frac{F - \text{median}(F)}{\max(\text{MAD}(F), \epsilon)}.$$

With $\text{MAD}(F) = \text{median}(|F - \text{median}(F)|)$ and machine epsilon ϵ , a block was flagged when $z_A > 4.5$ or $z_R > 4.5$. All its samples were marked, and any analysis window containing a marked sample was excluded. Trailing incomplete blocks were not screened.

Within each same-state run, complete non-overlapping 60-s windows used $\text{round}(60f_s)$ samples. Incomplete tails were discarded and windows crossing state boundaries were not formed. Candidates containing a timestamp interval greater than $1.5/f_s$ were excluded, with subsequent positions anchored to the original state run. The same procedure was used at 120 and 180 s.

Windows passing artifact and gap screening were assessed for variance stability using complete, non-overlapping 10-s subwindows of $\text{round}(10f_s)$ samples; at least two were required. With $v_k = \text{Var}(x_k)$ calculated using $\text{ddof} = 0$, the index was

$$S = \frac{\text{IQR}(v)}{\max(\text{median}(v), \epsilon)},$$

$$\text{IQR}(v) = Q_{0.75}(v) - Q_{0.25}(v).$$

Acceptance required $S \leq 0.5$. This screen supported local quasi-stationarity through variance stability, rather than establishing full stationarity. At 60 s, variance screening retained 596 Awake and 305 Drowsy windows from 623 and 328 passing artifact/gap screening. Final cohorts totaled 901, 405, and 238 windows at 60, 120, and 180 s (Table S2).

The 30-s cohort comprised two consecutive halves of each accepted 60-s parent window. These halves inherited parent QC and were not independently variance-screened, yielding 1,192 Awake and 610 Drowsy windows (1,802 total). LLE fit acceptance was applied separately (Section S7); nominal retention was 872/901 (96.78%). The complete metric-specific counts are in Table S2 and the preprocessing/retention illustration in Fig. S1.

Table S2: QC stages and metric-specific retained-window counts.

Duration (s)	State	Post-SQI/gap	Variance-pass / final non-LLE	LLE-valid	Variance-pass fraction	LLE-pass fraction
30	Awake	Not applicable	1,192	1,089	Not applicable	1089/1192 (91.36%)
30	Drowsy	Not applicable	610	549	Not applicable	549/610 (90.00%)
60	Awake	623	596	579	596/623 (95.67%)	579/596 (97.15%)
60	Drowsy	328	305	293	305/328 (92.99%)	293/305 (96.07%)
120	Awake	284	269	264	269/284 (94.72%)	264/269 (98.14%)
120	Drowsy	148	136	135	136/148 (91.89%)	135/136 (99.26%)
180	Awake	168	161	160	161/168 (95.83%)	160/161 (99.38%)
180	Drowsy	86	77	77	77/86 (89.53%)	77/77 (100.00%)

Variance-pass fractions divide the final non-LLE count by the post-SQI/gap count. LLE-pass fractions divide accepted fits by the final non-LLE count. SQI denotes amplitude/roughness signal-quality screening.

At 30 s, post-SQI/gap and independent variance-pass stages are not applicable because parent QC is inherited. Cohorts at different durations share source recordings.

S3 Frozen Nominal Analysis Configuration

The common nominal reconstruction and metric-specific settings are summarized in Table S3A. The same embedding was used in both states. Windows were computational units and recording sessions were primary inferential units. Correction families are listed in Table S3B; they are kept separate across analyses and configurations.

Table S3: Nominal configuration and statistical correction families.

Panel A. Nominal estimator configuration

Component	Configuration
Input and windowing	Processed PPG; 60-s non-overlapping windows; 596 Awake + 305 Drowsy = 901.
Embedding	$\tau = 0.16$ s; $m = 8$; $\tau_{\text{samples}} = \text{round}(\tau f_s)$ (4/8 samples at 25/50 Hz).
Simplex Projection	$k = m + 1 = 9$; Euclidean distance; normalized exponential weights; leave-one-out prediction; temporal exclusion $W = 1.0$ s; 18 physical-time horizons from 0.04 to 4.00 s, converted using $\text{round}(T_p f_s)$. Mean CC and Mean NRMSE are the means over horizons within each window.
RQA	Euclidean distance; per-window threshold targeting $RR = 0.02$; exclusion $W = (m - 1)\tau_{\text{samples}}$; $l_{\min} = v_{\min} = 2$. Outputs: DET, L_{mean} , LAM, TT. DET describes diagonal recurrence-line organization.
Rosenstein LLE	Nearest admissible Euclidean neighbor; spectral-mean-period temporal exclusion; maximum follow time 5 s; fit interval 0.80–1.30 s; ≥ 50 initial pairs; ≥ 30 pairs per fit point; ≥ 3 fit points; finite fit; $R^2 \geq 0.90$. Units: s^{-1} .
PPS	$M = 39$ surrogates per tested window; tested noisy pseudoperiodic null; two-sided finite-sample rank test.
Aggregation and effects	Session-state medians of valid window metrics; paired $\Delta_i = \text{Drowsy}_i - \text{Awake}_i$; report the median paired difference. Simplex aggregation follows the within-window horizon means above.

Table S3 (continued)

Component	Configuration
Primary statistics	Two-sided Wilcoxon signed-rank test; matched-pairs rank-biserial correlation; 20,000 paired-session bootstrap resamples; percentile 95% CI for the median paired difference; BH-FDR once across the seven nominal 60-s metrics.
Panel B. Statistical families	
Analysis family	Multiplicity reporting
Primary RQ2	BH across 7 nominal 60-s metrics, once
Label sensitivity	BH across 4 headline metrics within each alternative rule
Delay/dimension sensitivity	BH across 4 headline metrics within each setting
RQA RR/Theiler sensitivity	Raw p only
LLE sensitivity	Raw p only
Window-length robustness	Raw p only
Repeated-session sensitivity	BH across 7 metrics per hypothetical matching
Symbolic dynamics	BH across 3 metrics per duration
PPS window-level rank tests	Unadjusted
PPS session-level gap family	BH across 12 tests: 6 metrics $\times$ 2 states

The seven primary metrics are Mean CC, Mean NRMSE, DET, L_{mean} , LAM, TT, and LLE. The four headline metrics are Mean CC, Mean NRMSE, DET, and LLE. Benjamini–Hochberg adjustment controls the false discovery rate within each specified family; families are not pooled across settings.

The PPS session-gap family contains the six non-LLE metrics in each state. Nominal reference rows in sensitivity tables retain the primary seven-metric family.

S4 Phase-Space Reconstruction: AMI and FNN

Average Mutual Information (AMI) was calculated separately for each processed 60-s window using KSG estimator 1 with $k = 3$, Chebyshev distance in joint space, and natural-log units (nats). Candidate delays ranged from one sample to $\text{round}(f_s \times 1.0 \text{ s})$ samples. Selection used the first strict interior local minimum without endpoint fallback; all 901 windows yielded a selected delay (Table S4A).

Median selected delays were 0.160 s for Awake (Q25–Q75: 0.120–0.260 s) and 0.120 s for Drowsy (0.120–0.160 s). The median of 20 session medians supported the operational common delay $\tau = 0.16 \text{ s}$; 14/20 session medians were within $0.16 \pm 0.04 \text{ s}$. This delay was shared across states and was not optimized for a state contrast.

False Nearest Neighbors (FNN) were evaluated over dimensions 1–10 using Euclidean nearest neighbors and Kennel-style criteria. A neighbor was false if additional-coordinate separation divided by the m -dimensional distance R_m exceeded 15, or full $(m + 1)$ -dimensional distance divided by signal standard deviation exceeded 2. Self-matches and zero/near-zero distances were excluded; this diagnostic used no temporal exclusion.

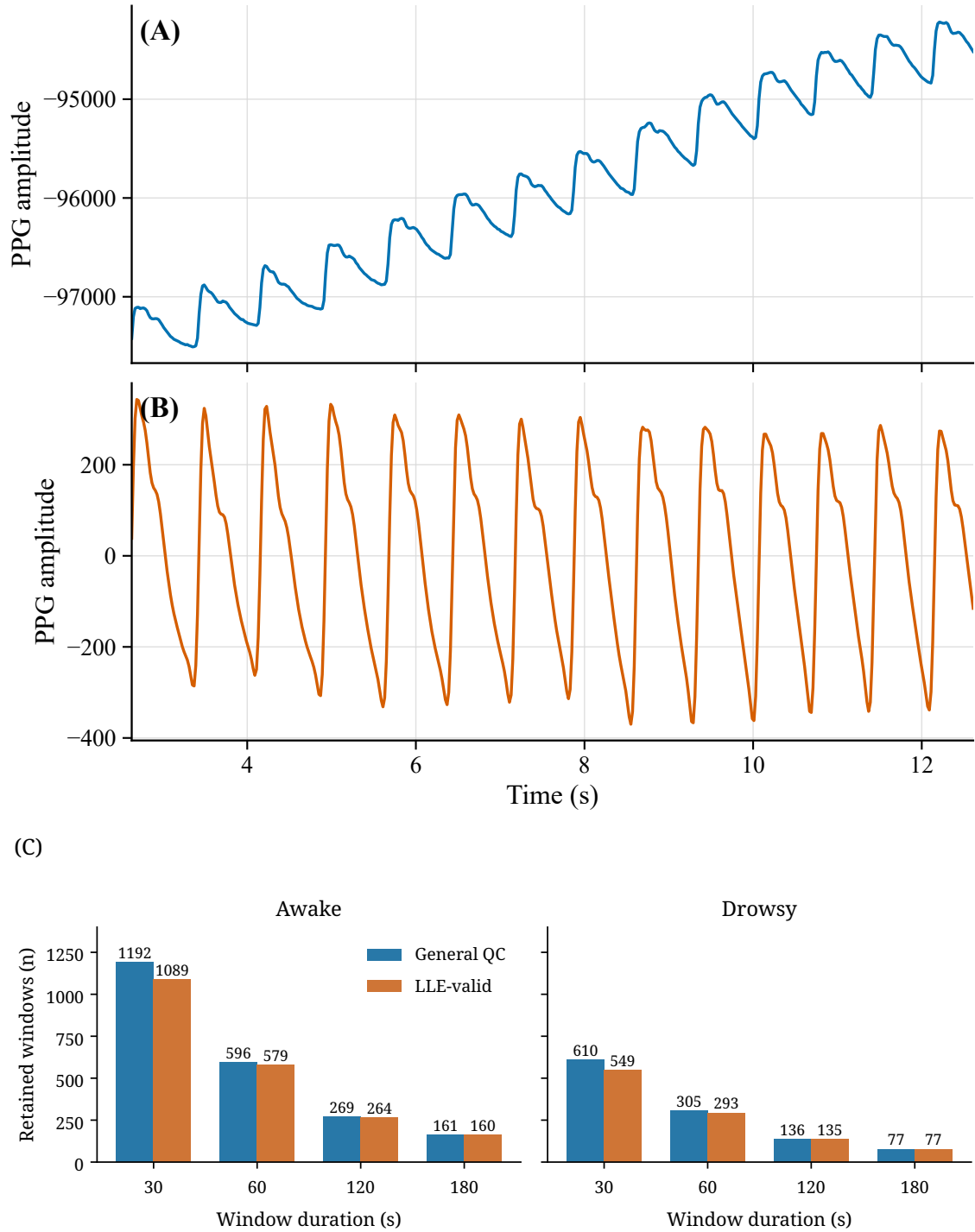


Figure S1: Preprocessing and retained-window counts. (A) Sign-inverted raw PPG and (B) processed PPG after the 0.5–8 Hz zero-phase Butterworth filter. (C) General-QC and LLE-valid window counts by state and duration, as listed in Table S2. The 30-s windows are consecutive halves of accepted 60-s parents and inherit their QC.

At the nominal delay, median FNN across 20 session curves was 0.900%, 0.886%, and 1.008% at $m = 7, 8, 9$, respectively (Table S4B). The common dimension $m = 8$ was selected from this low-FNN region. At $m = 8$, delays 0.12, 0.16, and 0.20 s yielded median FNN of 0.593%, 0.886%, and 1.284%, with 19/20, 11/20, and 7/20 session curves below 1% (Table S4C). Neither parameter was treated as uniquely optimal, and the nominal setting did not require every session to have FNN below 1%. These diagnostics describe operational reconstruction behavior (Fig. S2); inferential sensitivity is addressed in Section S12.1.

Table S4: AMI and FNN evidence for the common reconstruction.

Panel A. AMI-selected delays

State	Windows	Median τ (s)	Q25 (s)	Q75 (s)	Successful selections
Awake	596	0.160	0.120	0.260	596/596
Drowsy	305	0.120	0.120	0.160	305/305

Panel B. FNN by dimension at $\tau = 0.16$ s

Dimension	Median FNN (%)	Q25 (%)	Q75 (%)
7	0.900	0.679	1.155
8	0.886	0.613	1.132
9	1.008	0.666	1.298

Panel C. Delay sensitivity at $m = 8$

τ (s)	Median FNN (%)	Session curves <1%
0.12	0.593	19/20
0.16	0.886	11/20
0.20	1.284	7/20

Panel A summarizes selected delays across windows within each state; successful selections are relative to eligible windows. Panels B/C summarize session curves formed by median FNN across windows at each dimension. Their medians/quartiles and below-threshold counts are across 20 recording sessions. FNN is reported as a percentage.

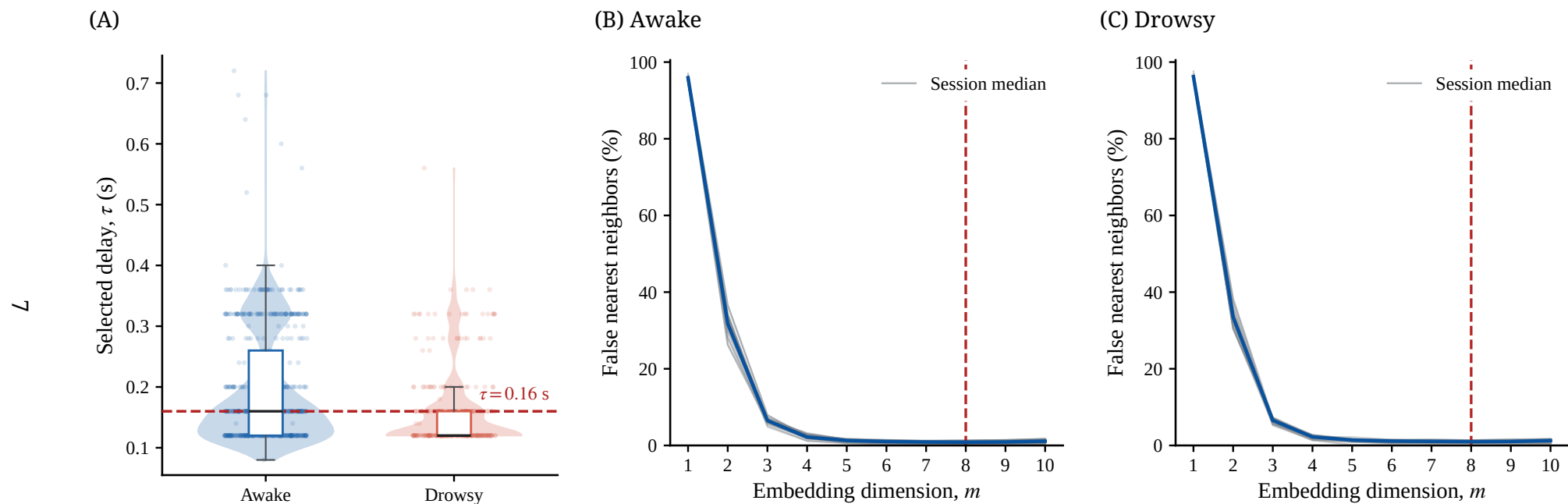


Figure S2: Reconstruction evidence from processed 60-s PPG. (A) AMI-selected delays, with the common $\tau = 0.16$ s. (B) Awake and (C) Drowsy FNN curves, with $m = 8$ in a low-FNN region. The same reconstruction was used for both states. These diagnostics support an operational choice rather than a universal optimum.

S5 Simplex Projection

Finite-horizon forecastability was assessed in the 901 retained windows. Each waveform was normalized as $u = (x - \bar{x})/\sigma_x$, using the whole-window population standard deviation (ddof = 0). The common reconstruction used $m = 8$, $\tau = 0.16$ s, and $d = \text{round}(\tau f_s)$ samples.

Forward delay vectors were $\mathbf{z}_i = [u_i, u_{i+d}, \dots, u_{i+(m-1)d}]$, with final-coordinate index $t_i = i + (m-1)d$. At horizon T_p , offset $h = \text{round}(T_p f_s)$ defined target u_{t_i+h} . Query and neighbor endpoints were restricted separately at each horizon so that vectors and targets remained within the same window. No cross-window prediction library was used.

Leave-one-out prediction used $k = m + 1 = 9$ nearest admissible Euclidean neighbors. Self-matches were excluded, and $|t_j - t_i| > \text{round}(f_s W)$ was required with $W = 1.0$ s. For neighbor distances D_j and smallest distance D_1 , weights and predictions were

$$w_j = \frac{\exp(-D_j/D_1)}{\sum_{r=1}^9 \exp(-D_r/D_1)},$$

$$\hat{u}_{t_i+h} = \sum_{j=1}^9 w_j u_{t_j+h}.$$

If selected distances were no greater than machine epsilon, those neighbors shared the full weight equally and other selected neighbors received zero weight. Predictions required all nine admissible neighbors. For valid observed/predicted targets, CC and NRMSE were

$$\text{CC} = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2 \sum_i (\hat{y}_i - \bar{\hat{y}})^2}},$$

$$\text{NRMSE} = \frac{\sqrt{n^{-1} \sum_i (y_i - \hat{y}_i)^2}}{\sigma_u}.$$

Here, n is the number of valid prediction pairs and σ_u is the population standard deviation of the entire normalized window, equal to one up to numerical precision. It was not recomputed from a horizon-specific target subset. Higher CC and lower NRMSE indicate greater forecastability. Mean CC and Mean NRMSE were averaged across 18 horizons within each window, then summarized by the median of valid windows within each session and state, as reported in the manuscript. The 901 windows yielded 16,218 horizon-level outputs.

Table S5A lists physical horizons and integer offsets at nominal sampling rates. Temporal-exclusion calibration tested $W \in \{0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.5, 2.0\}$ s. Adjacent-setting summaries pooled 120 session–state–duration combinations (20 sessions, two states, 60/120/180 s) and described absolute calibration changes, not state effects. The largest transition occurred from 0.6 to 0.8 s; 1.0 s was the first sustained stable setting after it (Table S5B). Horizon behavior is illustrated in Fig. S3.

Table S5: Simplex horizons and temporal-exclusion calibration.

Panel A. Physical horizons and sample offsets		
Horizon (s)	25-Hz offset (samples)	50-Hz offset (samples)
0.04	1	2
0.08	2	4
0.12	3	6
0.16	4	8
0.20	5	10
0.28	7	14
0.40	10	20

Table S5 (continued)

Horizon (s)	25-Hz offset (samples)	50-Hz offset (samples)
0.60	15	30
0.80	20	40
1.00	25	50
1.20	30	60
1.60	40	80
2.00	50	100
2.40	60	120
2.80	70	140
3.20	80	160
3.60	90	180
4.00	100	200

Panel B. Adjacent-setting calibration

Adjacent W (s)	Median absolute Δ CC	Median absolute Δ NRMSE	Minimum support	Stability /selection
0.0 $\rightarrow$ 0.2	0.003550	0.003370	1.00	Stable; before transition
0.2 $\rightarrow$ 0.4	2.204×10^{-6}	2.213×10^{-6}	1.00	Stable; before transition
0.4 $\rightarrow$ 0.6	0.002137	0.002059	1.00	Stable; before transition
0.6 $\rightarrow$ 0.8	0.020254	0.015614	1.00	Largest transition; unstable
0.8 $\rightarrow$ 1.0	0.003605	0.003260	1.00	Selected: W=1.0 s; sustained
1.0 $\rightarrow$ 1.5	0.008894	0.007937	1.00	Stable; sustained
1.5 $\rightarrow$ 2.0	0.007034	0.006442	1.00	Stable; sustained

Panel A offsets illustrate 25/50 Hz; actual offsets use estimated session f_s . Panel B gives median absolute adjacent-setting changes over 120 combinations, with support defined as the minimum valid-prediction fraction.

Stability required both change rates, normalized to their respective largest rate, to be ≤ 0.25 and minimum support ≥ 0.95 . Change rate is absolute change divided by the interval in W . Sustained stability required all subsequent pairs to pass; the selected upper setting was 1.0 s. These calibration quantities are not state-effect tests.

S6 Recurrence Quantification Analysis

RQA used Euclidean distances in the common reconstructed state space. A per-window threshold ε targeted $RR = 0.02$, with recurrence defined by distance $\leq \varepsilon$. The line of identity and pairs with $|i-j| \leq W$ were excluded, where $W = (m-1)d$: 28 samples at approximately 25 Hz and 56 at 50 Hz (approximately 1.12 s). Density calibration counted eligible upper-triangle distances once. At a tied boundary, attainable counts below/at the boundary were compared with the target; the closer RR was selected, with lower RR on equal error. Equal distances were included or excluded together.

Let C_{upper} be eligible upper-triangle recurrent points and $P_d(l)$ count maximal diagonal runs of length l . With $l_{\min} = 2$, diagonal recurrence-line organization and mean qualifying diagonal length were

$$\text{DET} = \frac{\sum_{l \geq 2} l P_d(l)}{C_{\text{upper}}},$$

$$L_{\text{mean}} = \frac{\sum_{l \geq 2} l P_d(l)}{\sum_{l \geq 2} P_d(l)}.$$

Vertical runs were counted column-wise in the full symmetric recurrence matrix after the same mask. For recurrent-point count C_{full} and maximal vertical-run counts $P_v(v)$, with $v_{\text{min}} = 2$, laminarity and trapping time were

$$\text{LAM} = \frac{\sum_{v \geq 2} v P_v(v)}{C_{\text{full}}},$$

$$\text{TT} = \frac{\sum_{v \geq 2} v P_v(v)}{\sum_{v \geq 2} P_v(v)}.$$

DET/LAM denominators included all recurrent points in their counting domains, including runs shorter than two. L_{mean} and TT averaged qualifying runs in sample-index line lengths, not seconds. The implementation returned zero when no qualifying runs were present, and zero fractions if the recurrence count was zero. DET describes diagonal recurrence-line organization; LAM/TT describe vertical organization, without a direct physical-determinism or autonomic interpretation. Fig. S4 illustrates recurrence geometry; nominal effects are in Table S7 and estimator sensitivity in Section S12.2.

S7 Rosenstein Largest Lyapunov Exponent

The Rosenstein estimator assessed finite-window local trajectory divergence using the common embedding. Its temporal exclusion was based on the spectral mean period of the demeaned processed waveform. Power was squared Fourier magnitude in a one-sided spectrum, excluding DC and retaining positive frequencies through Nyquist without an additional frequency mask. For power P_k at f_k ,

$$\bar{f} = \frac{\sum_{k: f_k > 0} f_k P_k}{\sum_{k: f_k > 0} P_k},$$

$$T = \bar{f}^{-1},$$

$$W = \max\{1, \text{round}(f_s T)\}.$$

The first nearest Euclidean neighbor satisfying $|j - i| > W$ was selected. Distances $\leq \delta = 10\epsilon_{\text{machine}} \max\{\text{std}(\mathbf{Z}), 1\} \sqrt{m}$ were then rejected without replacing the neighbor; $\text{std}(\mathbf{Z})$ is the population standard deviation across the state matrix. Admissible pairs were followed for at most 5 s. Pairs with in-window evolved endpoints and finite distance above δ contributed to the mean log-distance curve,

$$D(t) = \frac{1}{n_t} \sum_{j \in \mathcal{P}_t} \ln \|\mathbf{z}_{j+\ell} - \mathbf{z}_{\nu(j)+\ell}\|,$$

$$t = \ell / f_s.$$

Here, $\nu(j)$ identifies the selected neighbor, $\mathcal{P}_t$ is the valid pair set at lag ℓ , and n_t is its size. Linear regression against time in seconds over inclusive 0.80–1.30 s gave the slope in s^{-1} (Fig. S5). Acceptance required at least 50 initial pairs, 30 pairs per fit point, three finite fit points, a finite slope, and $R^2 \geq 0.90$.

Acceptance retained 872/901 input windows (96.78%): 579 Awake and 293 Drowsy. Across all input windows, minimum initial pair count was 1,471, minimum pair count at a fit point

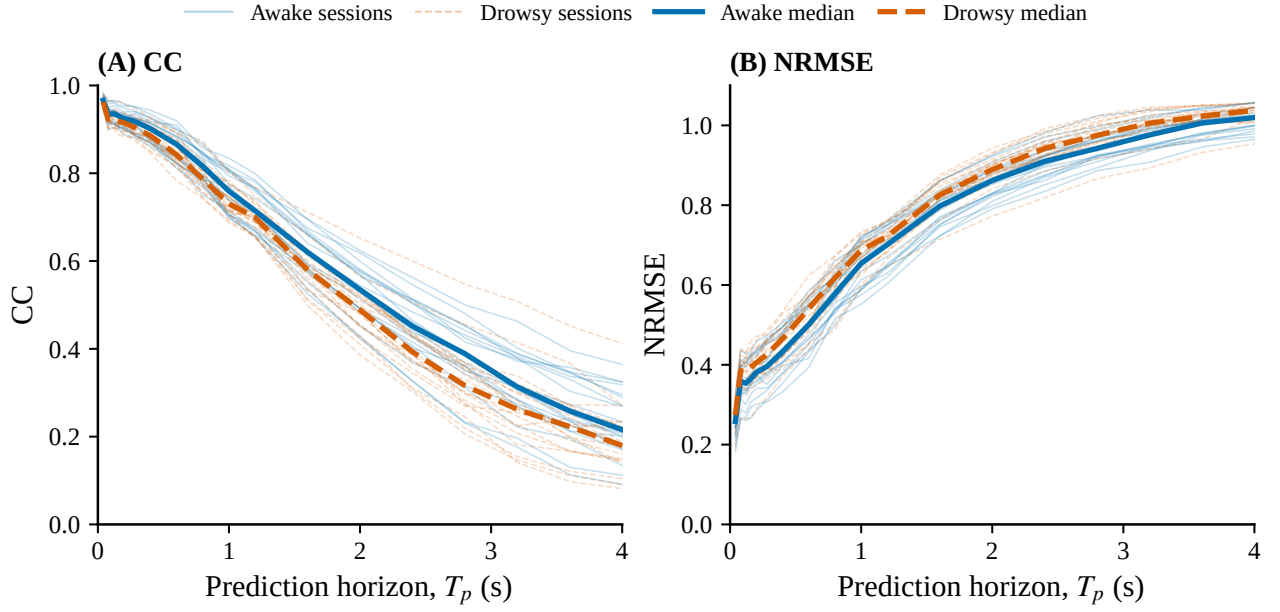


Figure S3: Simplex horizon behavior in processed 60-s PPG. (A) CC and (B) NRMSE across 18 physical-time horizons from 0.04 to 4.00 s, using the common reconstruction and 1.0-s temporal exclusion. Session traces and state medians show finite-horizon forecastability; paired state effects are in Table S7.

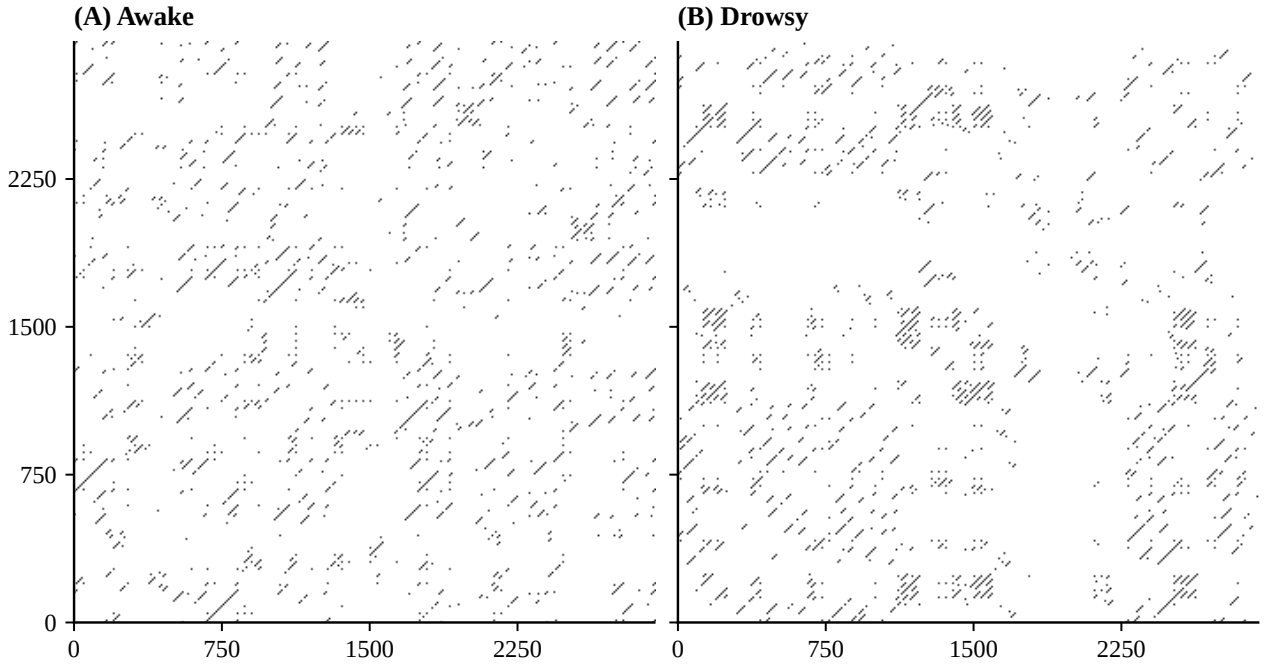


Figure S4: Example Awake and Drowsy recurrence plots from recording session 01, using processed 60-s windows, the common embedding, $RR = 0.02$, and exclusion $W = (m - 1)d$. Diagonal runs contribute to DET and L_{mean} ; vertical runs to LAM and TT. The plots illustrate recurrence geometry rather than inferential state effects.

was 1,366, and median fit R^2 was 0.973. Pair support therefore exceeded the operational minima throughout. The 29 rejected windows (17 Awake, 12 Drowsy) failed R^2 , rather than pair-support criteria. These diagnostics include rejected fits; state comparisons use accepted fits only. LLE is a finite-data divergence estimate, not evidence establishing deterministic chaos. Nominal effects are in Table S7; perturbations are in Section S12.3.

S8 Pseudoperiodic Surrogate Testing

Pseudoperiodic surrogates (PPS) tested a noisy pseudoperiodic null. Processed waveforms were forward-embedded using the common reconstruction and an initial index drawn uniformly from embedded states. For current state $\mathbf{s}$, candidate indices included all states with an observed successor, with probabilities

$$\Pr(j | \mathbf{s}) = \frac{\exp[-(\|\mathbf{z}_j - \mathbf{s}\| - D_{\min})/\rho]}{\sum_{r=0}^{N_e-2} \exp[-(\|\mathbf{z}_r - \mathbf{s}\| - D_{\min})/\rho]},$$

Here, N_e is the number of embedded states and $D_{\min}$ the smallest candidate distance; subtracting it stabilizes the exponential without changing normalized probabilities. After selecting j , the surrogate continued at observed successor $j + 1$. Self and temporally adjacent candidates were allowed; the last embedded state was excluded because it lacked a successor. The first coordinate of each visited state was emitted until the original window length was reached, without boundary wrapping.

Window-specific radius ρ maximized the mean number of maximal consecutive source-index runs of length at least two, using three trials at each of 21 candidate radii; ties used the first maximum. Each window was compared with $M = 39$ surrogates. The six non-LLE metrics shared an ensemble; LLE used a separate ensemble, retaining original-window exclusion but not imposing the original $R^2 \geq 0.90$ threshold on surrogate fits. LLE comparisons used the 579 Awake and 293 Drowsy original accepted fits.

For original value a and surrogate values $s_1, \dots, s_M$, the two-sided finite-sample test included ties in both tails:

$$p = \min \left\{ 1, \frac{2 \min(1 + \#\{s_b \leq a\}, 1 + \#\{s_b \geq a\})}{M + 1} \right\}.$$

The smallest attainable p was $2/40 = 0.05$, with rejection at $p \leq 0.05$ and no window-level adjustment. There were 24 exact original-surrogate ties across six-metric comparisons, none changing rejection at this threshold; included LLE comparisons had no ties.

Table S6A distinguishes pooled rejection percentages from median/IQR of 20 session-specific fractions. CC rejected in 374/596 Awake windows (62.75% pooled; 60.66% session median) and 229/305 Drowsy windows (75.08%; 71.01%). Expected original-minus-PPS directions were positive for CC, DET, L_{mean} , and LLE and negative for NRMSE, LAM, and TT. One Drowsy LAM rejection had the opposite direction; all others followed the expected direction.

For session-gap inference (Table S6B), each window contributed its original value minus the median of 39 surrogates. Session-state median gaps were summarized and tested across 20 sessions, with BH correction over 12 metric-state tests. Rejection fractions are descriptive; no LLE gap CI, p , or q is included in this family.

PPS rejection indicates departure from the tested noisy pseudoperiodic null. It neither establishes deterministic chaos nor excludes all stochastic or structured pseudoperiodic models. Within-state rejection fractions are not a paired Awake-Drowsy test. An illustrative waveform and rejection heatmaps are shown in Fig. S6.

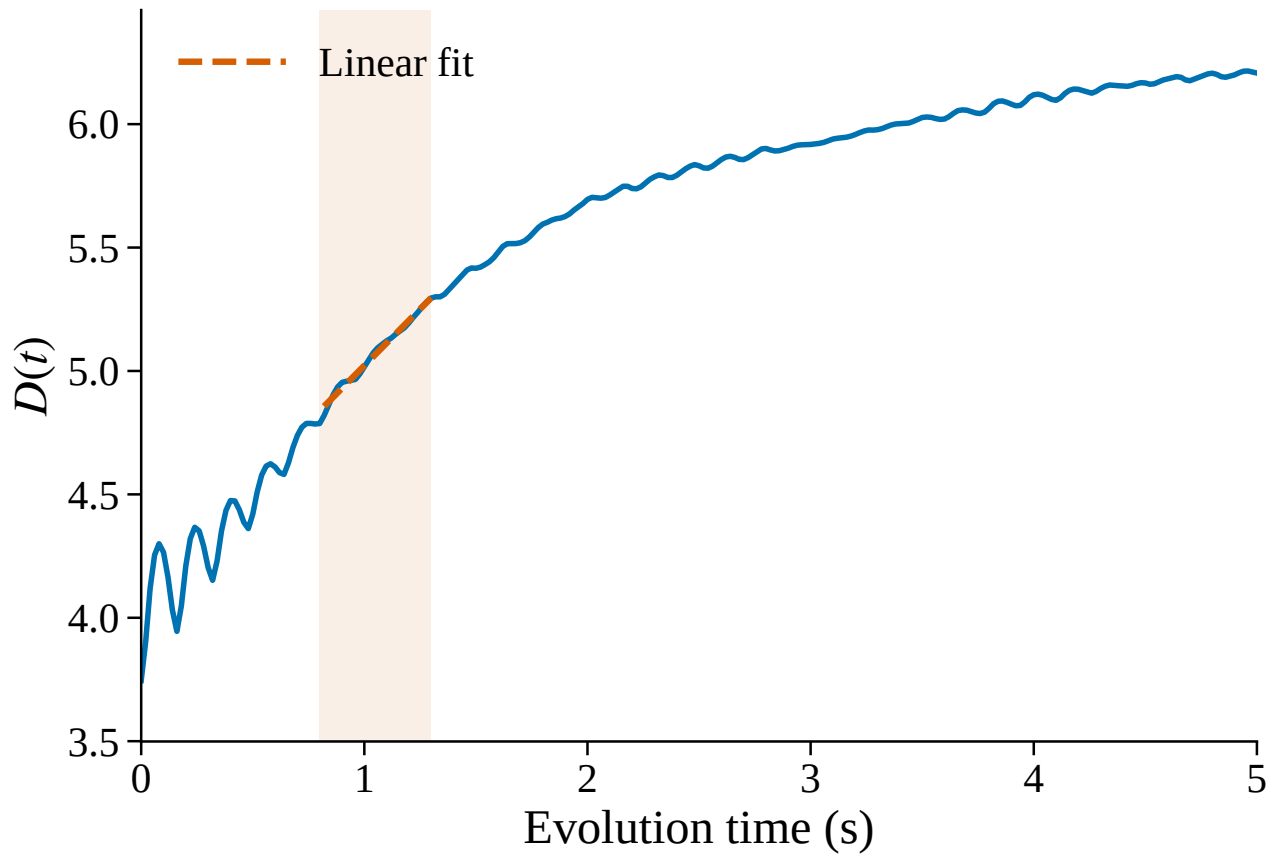


Figure S5: Example Rosenstein mean log-distance curve $D(t)$ and linear fit from a processed 60-s Awake window in recording session 01. The fit spans 0.80–1.30 s. Acceptance criteria are defined in Section S7. The slope is the finite-window LLE estimate in s^{-1} .

Table S6: PPS window rejection and session-level gap evidence.

Panel A. Window-level rejection summaries

Metric	State	Valid windows	Rejected n	Pooled (%)	Session median (%)	IQR (pp)	Expected-direction n	Opposite-direction n
Mean CC	Awake	596	374	62.75	60.66	16.98	374	0
Mean CC	Drowsy	305	229	75.08	71.01	23.31	229	0
Mean NRMSE	Awake	596	542	90.94	91.59	12.57	542	0
Mean NRMSE	Drowsy	305	291	95.41	100.00	7.27	291	0
DET	Awake	596	595	99.83	100.00	0.00	595	0
DET	Drowsy	305	305	100.00	100.00	0.00	305	0
L_{mean}	Awake	596	594	99.66	100.00	0.00	594	0
L_{mean}	Drowsy	305	305	100.00	100.00	0.00	305	0
LAM	Awake	596	537	90.10	95.50	16.36	537	0
LAM	Drowsy	305	273	89.51	92.38	18.59	272	1
TT	Awake	596	580	97.32	100.00	4.31	580	0
TT	Drowsy	305	295	96.72	100.00	7.28	295	0
LLE	Awake	579	545	94.13	96.30	7.28	545	0
LLE	Drowsy	293	287	97.95	100.00	0.76	287	0

Table S6 (continued)

Panel B. Session-level original-minus-PPS gaps

Metric	State	Sessions	Median gap	95% CI	Raw p	q_{BH}	r_{rb}	Session median rejection (%)
Mean CC	Awake	20	+0.1192	[0.1034, 0.1278]	1.90735×10^{-6}	1.90735×10^{-6}	+1.000	60.66
Mean CC	Drowsy	20	+0.1441	[0.1245, 0.1906]	1.90735×10^{-6}	1.90735×10^{-6}	+1.000	71.01
Mean NRMSE	Awake	20	-0.1321	[-0.1658, -0.1273]	1.90735×10^{-6}	1.90735×10^{-6}	-1.000	91.59
Mean NRMSE	Drowsy	20	-0.1557	[-0.1901, -0.1370]	1.90735×10^{-6}	1.90735×10^{-6}	-1.000	100.00
DET	Awake	20	+0.1833	[0.1727, 0.1939]	1.90735×10^{-6}	1.90735×10^{-6}	+1.000	100.00
DET	Drowsy	20	+0.1885	[0.1713, 0.1971]	1.90735×10^{-6}	1.90735×10^{-6}	+1.000	100.00
L_{mean}	Awake	20	+1.1528	[1.0947, 1.1891]	1.90735×10^{-6}	1.90735×10^{-6}	+1.000	100.00
L_{mean}	Drowsy	20	+1.1508	[1.1023, 1.1815]	1.90735×10^{-6}	1.90735×10^{-6}	+1.000	100.00
LAM	Awake	20	-0.1244	[-0.1411, -0.1130]	1.90735×10^{-6}	1.90735×10^{-6}	-1.000	95.50
LAM	Drowsy	20	-0.1266	[-0.1282, -0.1091]	1.90735×10^{-6}	1.90735×10^{-6}	-1.000	92.38
TT	Awake	20	-0.1139	[-0.1367, -0.0855]	1.90735×10^{-6}	1.90735×10^{-6}	-1.000	100.00
TT	Drowsy	20	-0.1188	[-0.1283, -0.1077]	1.90735×10^{-6}	1.90735×10^{-6}	-1.000	100.00

Panel A: CC/NRMSE are window-level horizon means. Pooled rejection equals rejected/valid windows $\times 100$; session median and IQR summarize 20 session-specific fractions scaled by 100. Expected signs are for original minus PPS: positive for CC, DET, L_{mean} , LLE and negative for NRMSE, LAM, TT. LLE uses 579/293 original accepted windows, versus 596/305 for other metrics.

Panel B: CI, Wilcoxon signed-rank p , r_{rb} , and BH-adjusted q describe session-state median original-minus-PPS gaps across 20 sessions. BH covers the 12 six-metric-state tests once. Rejection fractions are descriptive and LLE has no gap inference in this family. Gap/CI units are samples for L_{mean} /TT and dimensionless otherwise.

S9 Statistical Framework and Multiplicity

Each metric was summarized by the median of valid windows within each recording session and state. For Simplex, window metrics were first averaged over horizons (Section S5). Each of 20 sessions contributed a paired difference,

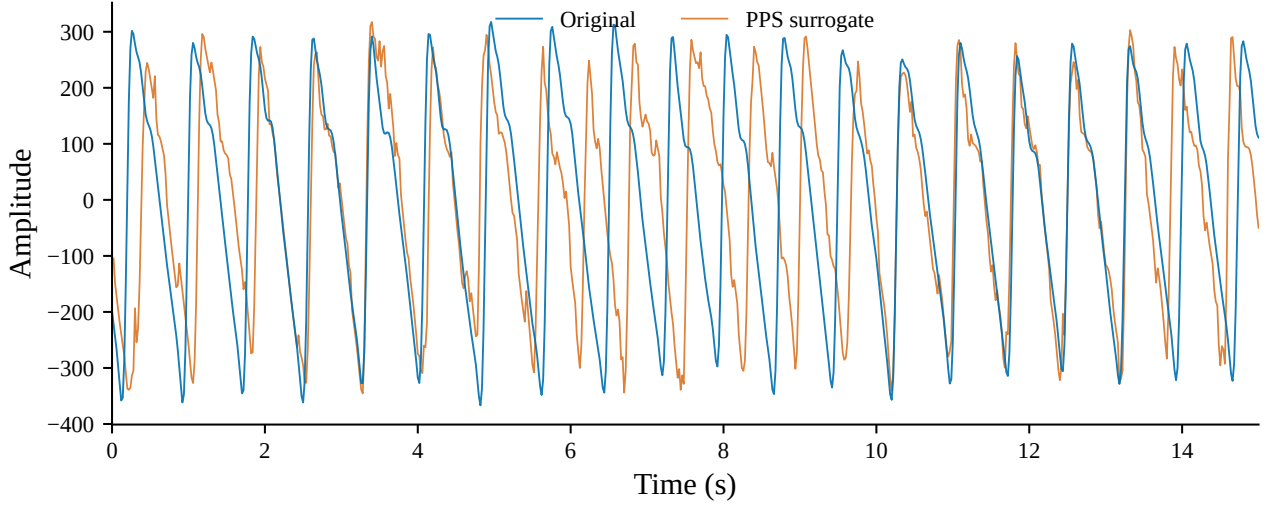
$$\begin{aligned}\Delta_i &= \text{Drowsy}_i - \text{Awake}_i, \\ \Delta &= \text{median}_i(\Delta_i).\end{aligned}$$

Two-sided Wilcoxon signed-rank tests removed zeros before ranking and assigned average ranks to equal absolute differences. Simplex treated absolute differences $\leq 10^{-12}$ as zero and used an exact conditional distribution; RQA/LLE removed exact zeros and used the automatic method without continuity correction. None of the seven nominal comparisons contained a zero difference.

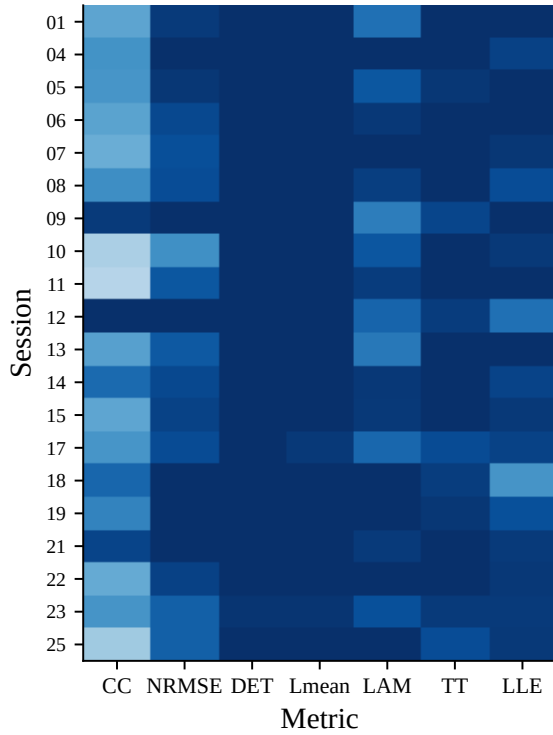
Matched-pairs rank-biserial correlation was $r_{rb} = (W_+ - W_-)/(W_+ + W_-)$, with positive/negative rank sums after zero removal. The 95% percentile bootstrap CI for the median paired effect used 20,000 resamples of the session-difference vector with replacement and the 2.5th/97.5th percentiles of resampled medians. Inferential and bootstrap units remained recording sessions (Section S1). CIs describe median-effect uncertainty; signed-rank tests depend on the signed rank distribution, so their conclusions need not agree exactly.

Primary BH-FDR adjustment covered the seven nominal 60-s metrics once, with support defined by $q_{BH} < 0.05$. Other families remain separate (Table S3B). In Table S7, Mean CC decreased and Mean NRMSE increased, indicating lower finite-horizon forecastability in Drowsy. DET decreased under nominal reconstruction and LLE decreased, describing lower diagonal recurrence organization and lower estimated local trajectory divergence. These four metrics met the BH7 criterion. L_{mean} was negative with a CI excluding zero but lacked BH support; LAM/TT also lacked support. These are complementary finite-window properties, rather than a common scale of chaos. All subsequent sensitivities reuse the same recordings and are supportive analyses, not independent replications.

(A)



(B) Awake



(C) Drowsy

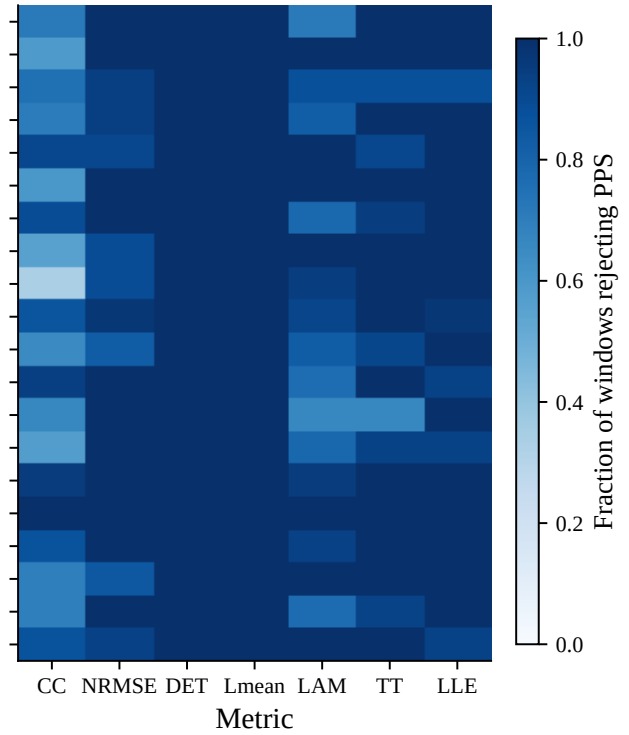


Figure S6: PPS illustration and rejection patterns. (A) A 15-s view of original processed PPG and an illustrative surrogate from recording session 01. (B) Awake and (C) Drowsy session-specific fractions rejecting the tested noisy pseudoperiodic null. Tests use 39 surrogates, tie-inclusive two-sided ranks, and unadjusted $p \leq 0.05$. LLE fractions use original accepted fits. The waveform is illustrative; heatmaps describe within-state rejection, not paired state effects.

Table S7: Complete primary 60-s statistics.

Metric	Paired n	Median Δ	95% CI	Raw p	q_{BH}	r_{rb}	Direction count
Mean CC	20	-0.0339	[-0.0595, -0.0043]	0.02148438	0.0376	-0.581	15/20 lower
Mean NRMSE	20	+0.0294	[0.0047, 0.0530]	0.00729561	0.0255	+0.667	15/20 higher
DET	20	-0.0186	[-0.0364, -0.0069]	0.02148438	0.0376	-0.581	15/20 lower
L_{mean}	20	-0.0630	[-0.1247, -0.0139]	0.05825806	0.0816	-0.486	16/20 lower
LAM	20	+0.0165	[-0.0044, 0.0635]	0.14290619	0.1667	+0.381	13/20 higher
TT	20	+0.0095	[-0.0034, 0.0160]	0.20244980	0.2024	+0.333	14/20 higher
LLE	20	-0.0376	[-0.0537, -0.0230]	0.00070763	0.0050	-0.810	16/20 lower

All comparisons use 20 paired recording sessions. General QC retained 596 Awake and 305 Drowsy windows (901 total); LLE used 579 and 293 accepted fits (872 total). Direction counts follow the reported median sign; no nominal differences were zero.

Δ is the median paired difference, not the difference of marginal state medians. Effect/CI units are samples for L_{mean} /TT, s^{-1} for LLE, and dimensionless otherwise. Raw p is shown to eight decimal places; manuscript effect, CI, r_{rb} , and q precision is preserved. q values belong to the primary BH7 family.

S10 Window-Length Robustness

Window-length sensitivity compared the four headline metrics at 30, 60, 120, and 180 s (Table S8; Fig. S7). Each duration had 20 paired sessions, with metric-specific eligible-window counts. Direction counts indicate decreases for Mean CC/DET/LLE and increases for Mean NRMSE. The paired-effect and CI conventions follow Section S9; this analysis reports raw p only.

All four median-effect directions were preserved across durations, but magnitudes and uncertainty differed. The 30-s estimates generally had greater uncertainty; longer windows generally reduced uncertainty while reducing eligible-window counts. The 180-s DET interval remained relatively wide. DET CIs included zero at 30 and 180 s, and the 180-s raw p was 0.053169. Direction preservation therefore did not imply uniform support. Within this dataset, 60 s provided a practical balance between observation duration and uncertainty, rather than a universal optimum.

Table S8: Window-length sensitivity of the four headline metrics.

Metric	Duration (s)	Awake windows	Drowsy windows	Paired n	Median Δ	95% CI	Raw p	r_{rb}	Direction count
Mean CC	30	1192	610	20	-0.0147	[-0.0712, -0.0039]	0.03276825	-0.543	15/20
Mean CC	60	596	305	20	-0.0339	[-0.0595, -0.0043]	0.02148438	-0.581	15/20
Mean CC	120	269	136	20	-0.0271	[-0.0490, -0.0156]	0.01361656	-0.619	16/20
Mean CC	180	161	77	20	-0.0428	[-0.0516, -0.0133]	0.00943565	-0.648	15/20
Mean NRMSE	30	1192	610	20	+0.0203	[+0.0009, +0.0554]	0.00943565	+0.648	14/20
Mean NRMSE	60	596	305	20	+0.0294	[0.0047, 0.0530]	0.00729561	+0.667	15/20
Mean NRMSE	120	269	136	20	+0.0321	[+0.0196, +0.0495]	0.00830841	+0.657	15/20
Mean NRMSE	180	161	77	20	+0.0321	[+0.0186, +0.0465]	0.00729561	+0.667	15/20
DET	30	1192	610	20	-0.0169	[-0.0318, +0.0002]	0.01531219	-0.610	14/20
DET	60	596	305	20	-0.0186	[-0.0364, -0.0069]	0.02148438	-0.581	15/20
DET	120	269	136	20	-0.0201	[-0.0399, -0.0134]	0.00638962	-0.676	15/20
DET	180	161	77	20	-0.0248	[-0.0416, +0.0008]	0.05316925	-0.495	14/20
LLE	30	1089	549	20	-0.0325	[-0.0549, -0.0066]	0.00315285	-0.724	17/20
LLE	60	579	293	20	-0.0376	[-0.0537, -0.0230]	0.00070763	-0.810	16/20
LLE	120	264	135	20	-0.0442	[-0.0591, -0.0216]	0.00232506	-0.743	18/20
LLE	180	160	77	20	-0.0498	[-0.0584, -0.0213]	0.00058556	-0.819	16/20

Counts are eligible metric-specific windows; LLE includes its fit screen (Section S7). Effects/CIs use s^{-1} for LLE and dimensionless units otherwise. Direction counts follow nominal expected signs. Nominal 60-s effect/CI/ r_{rb} values match Table S7; all duration p values are unadjusted.

S11 Label Sensitivity

P0 used primary recorded labels. T30/T60 excluded samples within $\pm 30/\pm 60$ s of transitions, while S3/S5 retained same-state episodes lasting at least 180/300 s. Windows were constructed and screened under each rule; cohorts need not be nested subsets of P0. General-QC retention is in Table S9A. P0/T30/T60/S3 retained 20 paired sessions and S5 retained 19; LLE additionally used fit acceptance.

All 16 alternative-rule median effects preserved the four headline directions and had $q_{\text{BH}} < 0.05$ within each rule's BH4 family (Table S9B; Fig. S8). Magnitudes differed. The T60 Mean CC CI $[-0.0561, +0.0059]$ and T30 LLE CI $[-0.0535, +0.0020]$ included zero, consistent with the CI/test distinction in Section S9. P0 remains the fixed primary reference, with BH7 rather than alternative-rule BH4 correction.

Table S9: Label definitions, retained cohorts, and four-metric effects.

Panel A. Rule definitions and general-QC retention

Rule	Definition	Awake windows	Drowsy windows	Total windows	Paired n
P0	Primary recorded labels	596	305	901	20
T30	Exclude samples within ± 30 s of transitions	586	274	860	20
T60	Exclude samples within ± 60 s of transitions	551	236	787	20
S3	Retain same-state episodes ≥ 180 s	609	291	900	20
S5	Retain same-state episodes ≥ 300 s	581	265	846	19

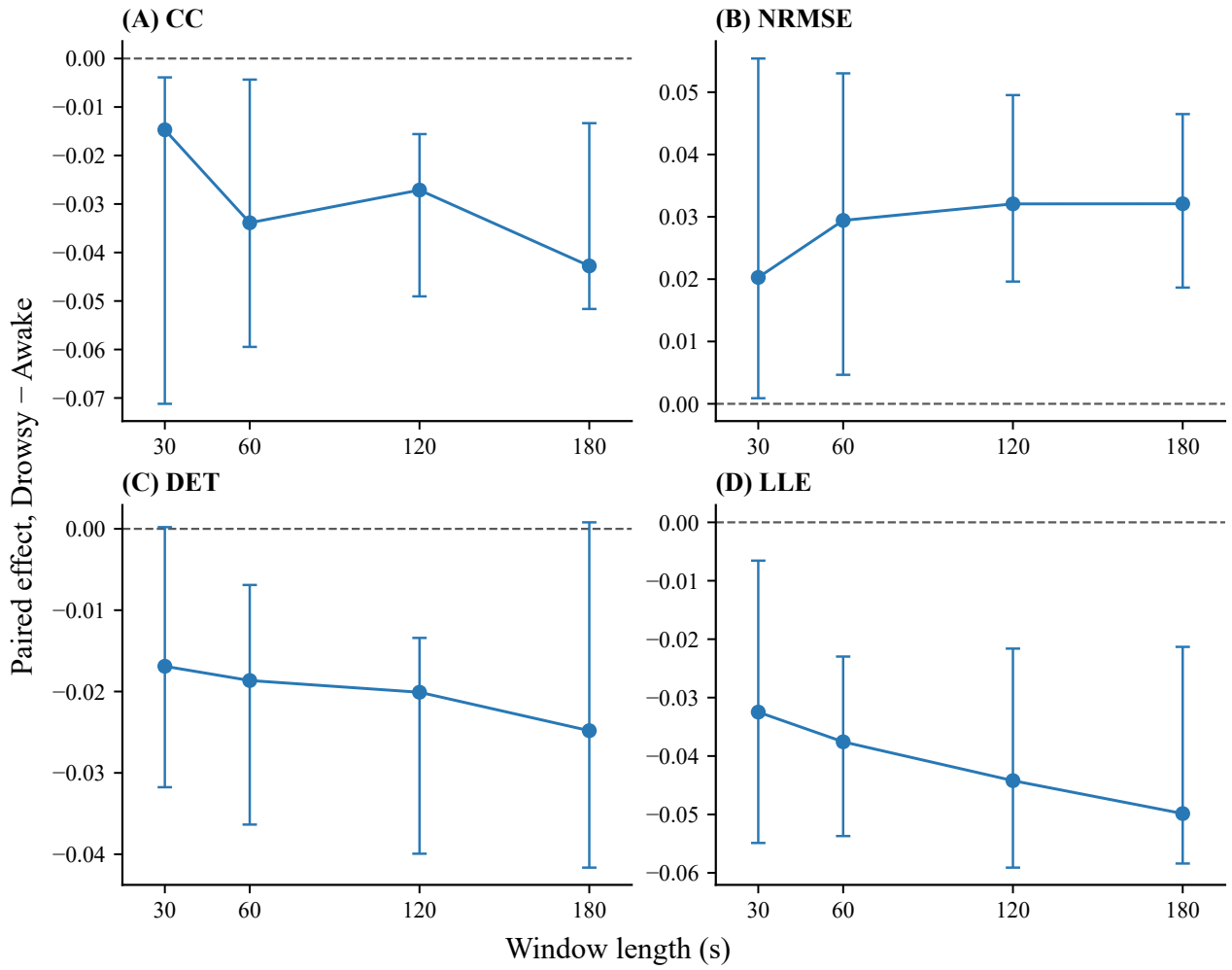


Figure S7: Window-length sensitivity. Median paired effects and 95% percentile bootstrap CIs for Mean CC, Mean NRMSE, DET, and LLE at 30, 60, 120, and 180 s. Median directions are preserved, while magnitude and uncertainty vary; DET intervals include zero at 30 and 180 s. Each duration has 20 paired sessions, with fewer eligible windows at longer durations and additional LLE fit screening (Table S8).

Table S9 (continued)

Panel B. Four-metric inference

Rule	Family	Metric	Paired n	Median Δ	95% CI	Raw p	q_{BH}	r_{rb}	Direction count
P0	Primary BH7	Mean CC	20	-0.0339	[-0.0595, -0.0043]	0.02148438	0.0376	-0.581	15/20
P0	Primary BH7	Mean NRMSE	20	+0.0294	[0.0047, 0.0530]	0.00729561	0.0255	+0.667	15/20
P0	Primary BH7	DET	20	-0.0186	[-0.0364, -0.0069]	0.02148438	0.0376	-0.581	15/20
P0	Primary BH7	LLE	20	-0.0376	[-0.0537, -0.0230]	0.00070763	0.0050	-0.810	16/20
T30	Sensitivity BH4	Mean CC	20	-0.0482	[-0.0689, -0.0152]	0.0048599	0.0097	-0.695	17/20
T30	Sensitivity BH4	Mean NRMSE	20	+0.0406	[+0.0190, +0.0570]	0.0031528	0.0097	+0.724	16/20
T30	Sensitivity BH4	DET	20	-0.0183	[-0.0332, -0.0011]	0.0083084	0.0111	-0.657	14/20
T30	Sensitivity BH4	LLE	20	-0.0322	[-0.0535, +0.0020]	0.021484	0.0215	-0.581	14/20
T60	Sensitivity BH4	Mean CC	20	-0.0251	[-0.0561, +0.0059]	0.039989	0.0400	-0.524	13/20
T60	Sensitivity BH4	Mean NRMSE	20	+0.0278	[+0.0027, +0.0479]	0.013617	0.0256	+0.619	14/20
T60	Sensitivity BH4	DET	20	-0.0167	[-0.0294, -0.0066]	0.019234	0.0256	-0.590	15/20
T60	Sensitivity BH4	LLE	20	-0.0393	[-0.0636, -0.0075]	0.0023251	0.0093	-0.743	15/20
S3	Sensitivity BH4	Mean CC	20	-0.0361	[-0.0548, -0.0033]	0.010689	0.0136	-0.638	14/20
S3	Sensitivity BH4	Mean NRMSE	20	+0.0357	[+0.0032, +0.0457]	0.0063896	0.0128	+0.676	15/20
S3	Sensitivity BH4	DET	20	-0.0189	[-0.0326, -0.0069]	0.013617	0.0136	-0.619	15/20
S3	Sensitivity BH4	LLE	20	-0.0344	[-0.0495, -0.0104]	0.0010166	0.0041	-0.790	15/20
S5	Sensitivity BH4	Mean CC	19	-0.0337	[-0.0495, -0.0077]	0.01236	0.0141	-0.642	14/19
S5	Sensitivity BH4	Mean NRMSE	19	+0.0367	[+0.0056, +0.0504]	0.0045776	0.0092	+0.716	16/19
S5	Sensitivity BH4	DET	19	-0.0213	[-0.0305, -0.0049]	0.014069	0.0141	-0.632	15/19
S5	Sensitivity BH4	LLE	19	-0.0321	[-0.0499, -0.0147]	0.0014114	0.0056	-0.789	14/19

Panel A retention precedes LLE fit screening. In Panel B the Family column distinguishes primary BH7 reference rows from BH4 sensitivity rows; correction is never pooled across rules.

Direction counts are decreases for Mean CC/DET/LLE and increases for Mean NRMSE. Effect/CI units are s^{-1} for LLE and dimensionless otherwise. Raw p retains source precision; alternative-rule q is shown to four decimals.

S12 Reconstruction and Estimator Sensitivity

One reconstruction or estimator setting was perturbed at a time, retaining nominal choices for other settings. These analyses describe dependence within the tested configurations.

S12.1 Embedding Delay and Dimension

Delay sensitivity used $\tau = 0.12, 0.16, 0.20$ s at $m = 8$; dimension sensitivity used $m = 7, 8, 9$ at $\tau = 0.16$ s (Table S10). Mean CC, Mean NRMSE, and LLE retained negative, positive, and negative median effects across delays. DET was delay-sensitive: its nominal -0.0186 became near-zero and positive at 0.12 s ($+0.0025$) and 0.20 s ($+0.0015$). Both non-nominal CIs spanned zero and lacked BH support.

All four median directions were preserved across $m = 7-9$, but magnitudes varied, including LLE. DET lacked BH4 support at $m = 7$ ($q = 0.053169$) and $m = 9$ ($q = 0.058258$), despite negative effects and CIs excluding zero. Some Mean CC/NRMSE CIs also spanned zero. Table S10 distinguishes nominal primary BH7 rows from alternative-setting BH4 rows; no correction is pooled across settings. DET is therefore not uniformly stable to embedding delay.

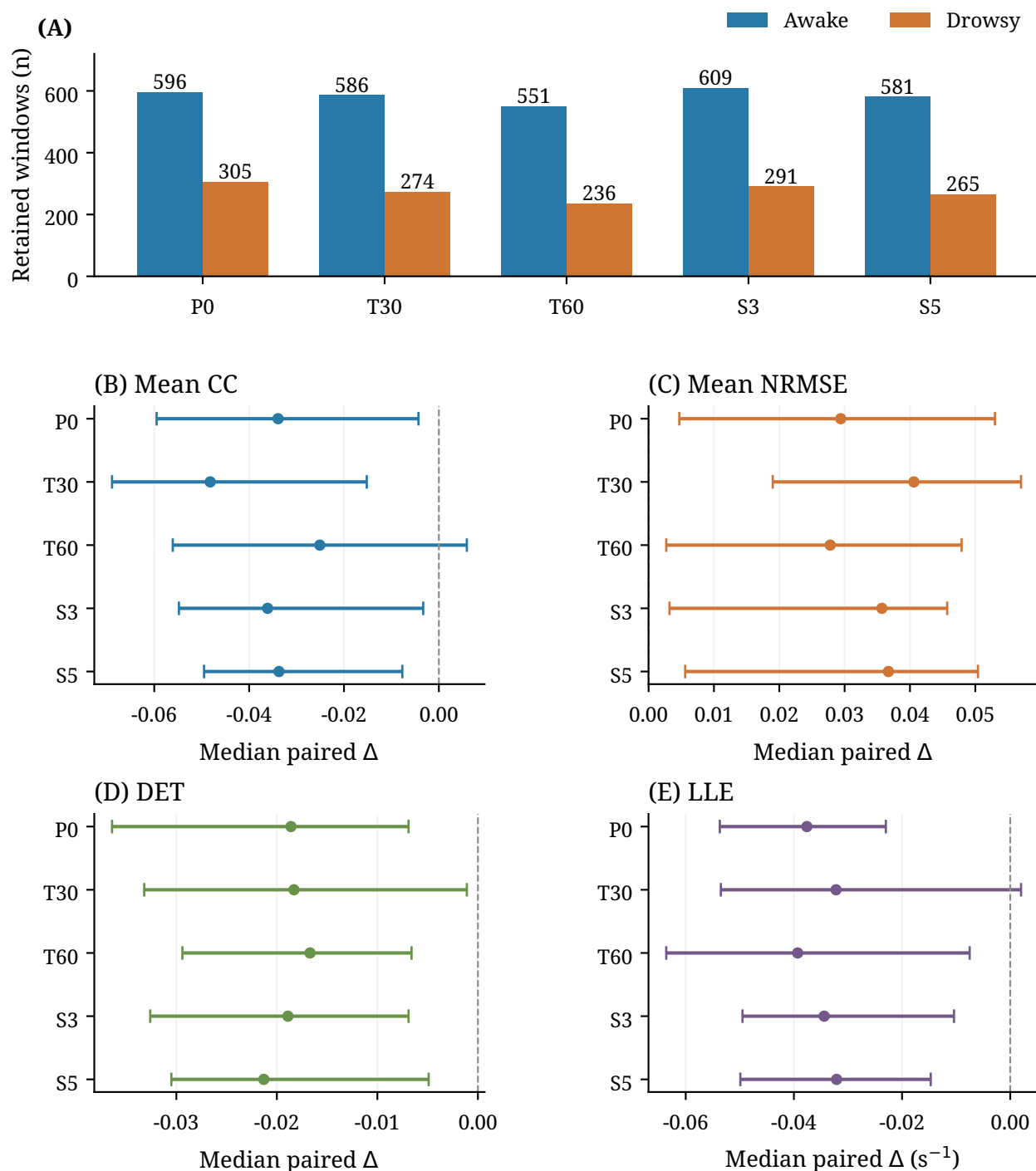


Figure S8: Label sensitivity. (A) Awake/Drowsy general-QC retention under P0/T30/T60/S3/S5. (B–E) Four-metric median paired effects and 95% percentile CIs, using Table S7 for P0 and alternative-rule summaries for sensitivities. Paired n is 20 except S5 (19). Median directions are preserved; T60 Mean CC and T30 LLE CIs include zero. Alternative-rule support uses BH4 within rule; P0 retains primary BH7.

Table S10: Reconstruction sensitivity with explicit correction families.

Panel A. Delay sensitivity at m=8

Setting	Family	Metric	Paired n	Median Δ	95% CI	Raw p	q_{BH}	r_{rb}	Direction count
0.12	Sensitivity BH4	Mean CC	20	-0.0317	[-0.0522, +0.0094]	0.036234	0.048312	-0.533	13/20
0.12	Sensitivity BH4	Mean NRMSE	20	+0.0321	[-0.0042, +0.0492]	0.029575	0.048312	+0.552	13/20
0.12	Sensitivity BH4	DET	20	+0.0025	[-0.0078, +0.0076]	0.784126	0.784126	+0.076	7/20
0.12	Sensitivity BH4	LLE	20	-0.0361	[-0.0567, +0.0042]	0.017181	0.048312	-0.600	14/20
0.16	Primary BH7	Mean CC	20	-0.0339	[-0.0595, -0.0043]	0.02148438	0.0376	-0.581	15/20
0.16	Primary BH7	Mean NRMSE	20	+0.0294	[0.0047, 0.0530]	0.00729561	0.0255	+0.667	15/20
0.16	Primary BH7	DET	20	-0.0186	[-0.0364, -0.0069]	0.02148438	0.0376	-0.581	15/20
0.16	Primary BH7	LLE	20	-0.0376	[-0.0537, -0.0230]	0.00070763	0.0050	-0.810	16/20
0.20	Sensitivity BH4	Mean CC	20	-0.0316	[-0.0529, -0.0008]	0.015312	0.020416	-0.610	14/20
0.20	Sensitivity BH4	Mean NRMSE	20	+0.0296	[+0.0056, +0.0476]	0.005581	0.011162	+0.686	16/20
0.20	Sensitivity BH4	DET	20	+0.0015	[-0.0099, +0.0148]	0.985435	0.985435	+0.010	10/20
0.20	Sensitivity BH4	LLE	20	-0.0464	[-0.0572, -0.0240]	0.000036	0.000145	-0.933	17/20

Table S10 (continued)

Panel B. Dimension sensitivity at delay 0.16 s

Setting	Family	Metric	Paired n	Median Δ	95% CI	Raw p	q_{BH}	r_{rb}	Direction count
7	Sensitivity BH4	Mean CC	20	-0.0307	[-0.0565, +0.0045]	0.026642	0.035522	-0.562	13/20
7	Sensitivity BH4	Mean NRMSE	20	+0.0377	[-0.0023, +0.0561]	0.023951	0.035522	+0.571	13/20
7	Sensitivity BH4	DET	20	-0.0145	[-0.0303, -0.0047]	0.053169	0.053169	-0.495	15/20
7	Sensitivity BH4	LLE	20	-0.0285	[-0.0549, -0.0076]	0.004860	0.019440	-0.695	16/20
8	Primary BH7	Mean CC	20	-0.0339	[-0.0595, -0.0043]	0.02148438	0.0376	-0.581	15/20
8	Primary BH7	Mean NRMSE	20	+0.0294	[0.0047, 0.0530]	0.00729561	0.0255	+0.667	15/20
8	Primary BH7	DET	20	-0.0186	[-0.0364, -0.0069]	0.02148438	0.0376	-0.581	15/20
8	Primary BH7	LLE	20	-0.0376	[-0.0537, -0.0230]	0.00070763	0.0050	-0.810	16/20
9	Sensitivity BH4	Mean CC	20	-0.0251	[-0.0492, +0.0055]	0.029575	0.039434	-0.552	13/20
9	Sensitivity BH4	Mean NRMSE	20	+0.0315	[+0.0043, +0.0483]	0.008308	0.016617	+0.657	15/20
9	Sensitivity BH4	DET	20	-0.0200	[-0.0321, -0.0052]	0.058258	0.058258	-0.486	15/20
9	Sensitivity BH4	LLE	20	-0.0247	[-0.0534, -0.0102]	0.000586	0.002342	-0.819	16/20

Panel A settings are delay in seconds; Panel B settings are dimension. Primary BH7 rows reproduce Table S7, while sensitivity BH4 rows use the four-metric family within each alternative setting.

Direction counts follow nominal expected signs (Mean CC/DET/LLE decrease; Mean NRMSE increase), including when aggregate DET reverses. Effects/CIs use manuscript precision and units; alternative p/q retains six-decimal source precision. This is a one-factor analysis, not a full factorial grid.

S12.2 RQA Estimator Settings

RQA sensitivity tested $RR = 0.01, 0.02, 0.03$ and Theiler multipliers 0.75, 1.00, 1.25 relative to nominal $W = (m - 1)d$. All settings retained 901 windows and 20 paired sessions. Table S11A reports available DET/LAM/TT inference with raw p only. DET remained negative across these settings, although at $RR=0.01$ its CI spanned zero and p was 0.082550. Direction stability under these RQA choices contrasts with delay-dependent sign reversal in Section S12.1.

LAM/TT remained secondary: all median effects were positive, all CIs spanned zero, and all raw p values exceeded 0.05. L_{mean} estimator sensitivity was not available. Metric interpretation follows Section S6.

S12.3 LLE Estimator Settings

LLE sensitivity tested fit intervals 0.60–1.10, 0.80–1.30 (nominal), and 1.00–1.50 s; R^2 thresholds 0.90/0.95; and Theiler multipliers 0.75/1.00/1.25 relative to its spectral-mean-period exclusion (Section S7). All settings retained 20 paired sessions. Every median effect was negative, every CI excluded zero, and raw p was below 0.05 (Table S11B). No BH correction was applied across estimator settings.

Fit intervals retained 802/872/889 of 901 input windows, respectively. Increasing R^2 from 0.90 to 0.95 retained 734 rather than 872 windows and reduced the median decrease. The 0.75/1.00 Theiler multipliers gave the same effect and retention; 1.25 retained 874 windows and changed magnitude. Direction preservation therefore coexists with magnitude and cohort changes; the estimator is not parameter-independent.

Table S11: RQA and LLE estimator sensitivity.

Panel A. RQA RR and Theiler settings

Factor	Setting	Metric	Valid windows	Paired n	Median Δ	95% CI	Raw p	r_{rb}	Direction count
RR	0.01	DET	901	20	-0.015214	[-0.033764, +0.002064]	0.082550	-0.447619	14/20
RR	0.01	LAM	901	20	0.014242	[-0.005819, +0.057725]	0.132727	0.390476	13/20
RR	0.01	TT	901	20	0.000343	[-0.000043, +0.004808]	0.214602	0.352941	10/20; 4 zero
RR	0.02	DET	901	20	-0.0186	[-0.0364, -0.0069]	0.021484	-0.581	15/20
RR	0.02	LAM	901	20	+0.0165	[-0.0044, 0.0635]	0.142906	+0.381	13/20
RR	0.02	TT	901	20	+0.0095	[-0.0034, 0.0160]	0.202450	+0.333	14/20
RR	0.03	DET	901	20	-0.017813	[-0.033859, -0.007863]	0.021484	-0.580952	15/20
RR	0.03	LAM	901	20	0.020281	[-0.005463, +0.051395]	0.082550	0.447619	13/20
RR	0.03	TT	901	20	0.022063	[-0.007639, +0.033264]	0.294252	0.276190	14/20
Theiler multiplier	0.75	DET	901	20	-0.019191	[-0.036836, -0.006809]	0.021484	-0.580952	15/20
Theiler multiplier	0.75	LAM	901	20	0.019785	[-0.003072, +0.062855]	0.123093	0.400000	13/20
Theiler multiplier	0.75	TT	901	20	0.009809	[-0.003114, +0.015962]	0.189348	0.342857	14/20
Theiler multiplier	1.00	DET	901	20	-0.0186	[-0.0364, -0.0069]	0.021484	-0.581	15/20
Theiler multiplier	1.00	LAM	901	20	+0.0165	[-0.0044, 0.0635]	0.142906	+0.381	13/20
Theiler multiplier	1.00	TT	901	20	+0.0095	[-0.0034, 0.0160]	0.202450	+0.333	14/20
Theiler multiplier	1.25	DET	901	20	-0.019033	[-0.037873, -0.007730]	0.017181	-0.600000	15/20
Theiler multiplier	1.25	LAM	901	20	0.019372	[-0.005161, +0.062616]	0.132727	0.390476	12/20
Theiler multiplier	1.25	TT	901	20	0.009844	[-0.002347, +0.017601]	0.176853	0.352381	14/20

Table S11 (continued)
Panel B. LLE fit, quality, and Theiler settings

Factor	Setting	Passing / input	Awake win- dows	Drowsy win- dows	Paired n	Median Δ	95% CI	Raw p	r_{rb}	Decrease count
Fit interval (s)	0.60–1.10	802/901	542	260	20	-0.034980	[-0.064607, -0.019121]	0.000261	-0.857143	18/20
Fit interval (s)	0.80–1.30	872/901	579	293	20	-0.0376	[-0.0537, -0.0230]	0.000708	-0.810	16/20
Fit interval (s)	1.00–1.50	889/901	588	301	20	-0.030357	[-0.045778, -0.003941]	0.008308	-0.657143	15/20
R^2 threshold	0.90	872/901	579	293	20	-0.0376	[-0.0537, -0.0230]	0.000708	-0.810	16/20
R^2 threshold	0.95	734/901	500	234	20	-0.030618	[-0.041701, -0.013987]	0.002325	-0.742857	16/20
Theiler multiplier	0.75	872/901	579	293	20	-0.037570	[-0.053698, -0.022976]	0.000708	-0.809524	16/20
Theiler multiplier	1.00	872/901	579	293	20	-0.0376	[-0.0537, -0.0230]	0.000708	-0.810	16/20
Theiler multiplier	1.25	874/901	580	294	20	-0.029763	[-0.056095, -0.016331]	0.002325	-0.742857	16/20

Both panels report raw p only, with no BH correction across settings. Paired n is 20 throughout. Panel A counts decreases for DET and increases for LAM/TT; RR=0.01 TT has four zero differences removed before rank calculations. DET/LAM are dimensionless and TT uses sample-index line lengths.

Panel B passing counts use the 901 input-window denominator; effects/CIs are in s^{-1} . Theiler multipliers use each estimator’s own nominal exclusion. Nominal effects/CIs/ r_{rb} reproduce Table S7 at manuscript precision; perturbation values retain six decimals.

S13 Unknown Repeated-Session Dependence

To assess the linkage limitation in Section S1, a hypothetical dependence analysis grouped 20 sessions into 10 two-session clusters using random perfect matchings. Sampling with replacement paired adjacent entries in random permutations, producing 100,000 sampled matchings and 99,990 unique matchings among 654,729,075 theoretically possible matchings. These clusters do not recover participant identity or establish actual session independence.

For each matching, cluster differences were arithmetic means of the two session differences; the effect was the median of 10 cluster differences. Conditional two-sided signed-rank tests discarded differences within 10^{-12} of zero and averaged ranks for absolute-value ties. BH correction covered seven metrics within each matching, without nested bootstrap. Table S12A gives matching-distribution percentiles, not bootstrap CIs.

All four headline median directions were preserved in every sampled matching. Fractions with $q_{BH} < 0.05$ were 19.36% (Mean CC), 27.82% (Mean NRMSE), 22.59% (DET), and 79.69% (LLE). Direction was more stable than support after reducing effective units to 10 hypothetical clusters; LLE retained the strongest support within this analysis. Frequencies describe the specified hypothetical matching design, not probabilities that the true result is significant. They do not provide participant-level inference. Cumulative 50,000/100,000-draw summaries were similar (Table S12B; Fig. S9), supporting Monte Carlo stability without resolving linkage.

Table S12: Hypothetical matching distributions and convergence.

Panel A. Hypothetical matching distribution, 100,000 draws

Metric	Nominal direction	Sign preserved (%)	$q_{BH} < 0.05$ (%)	Median matching Δ	2.5th–97.5th percentile range
Mean CC	Decrease	100.00	19.36	-0.0242	[-0.0359, -0.0147]
Mean NRMSE	Increase	100.00	27.82	+0.0263	[+0.0185, +0.0343]
DET	Decrease	100.00	22.59	-0.0181	[-0.0255, -0.0100]
L_{mean}	Decrease	100.00	2.41	-0.0657	[-0.1000, -0.0333]
LAM	Increase	99.63	2.94	+0.0227	[+0.0063, +0.0402]
TT	Increase	99.25	0.04	+0.0076	[+0.0012, +0.0137]
LLE	Decrease	100.00	79.69	-0.0356	[-0.0446, -0.0266]

Panel B. Cumulative Monte Carlo summaries

Draws	Metric	Sign preserved (%)	$q_{BH} < 0.05$ (%)	Median matching Δ	2.5th–97.5th percentile range
50000	Mean CC	100.00	19.44	-0.0242	[-0.0360, -0.0147]
50000	Mean NRMSE	100.00	27.94	+0.0263	[+0.0185, +0.0342]
50000	DET	100.00	22.69	-0.0181	[-0.0256, -0.0100]
50000	LLE	100.00	79.69	-0.0356	[-0.0446, -0.0266]
100000	Mean CC	100.00	19.36	-0.0242	[-0.0359, -0.0147]
100000	Mean NRMSE	100.00	27.82	+0.0263	[+0.0185, +0.0343]
100000	DET	100.00	22.59	-0.0181	[-0.0255, -0.0100]
100000	LLE	100.00	79.69	-0.0356	[-0.0446, -0.0266]

Sign preservation compares each matching’s median cluster effect with the nominal direction. q support uses BH7 within matching. Ranges are 2.5th–97.5th percentiles of the hypothetical matching distribution, not bootstrap CIs or participant-effect intervals. Units follow Table S7.

The 50k/100k rows are cumulative prefixes of the same sequence, not independent runs. No participant mapping is inferred.

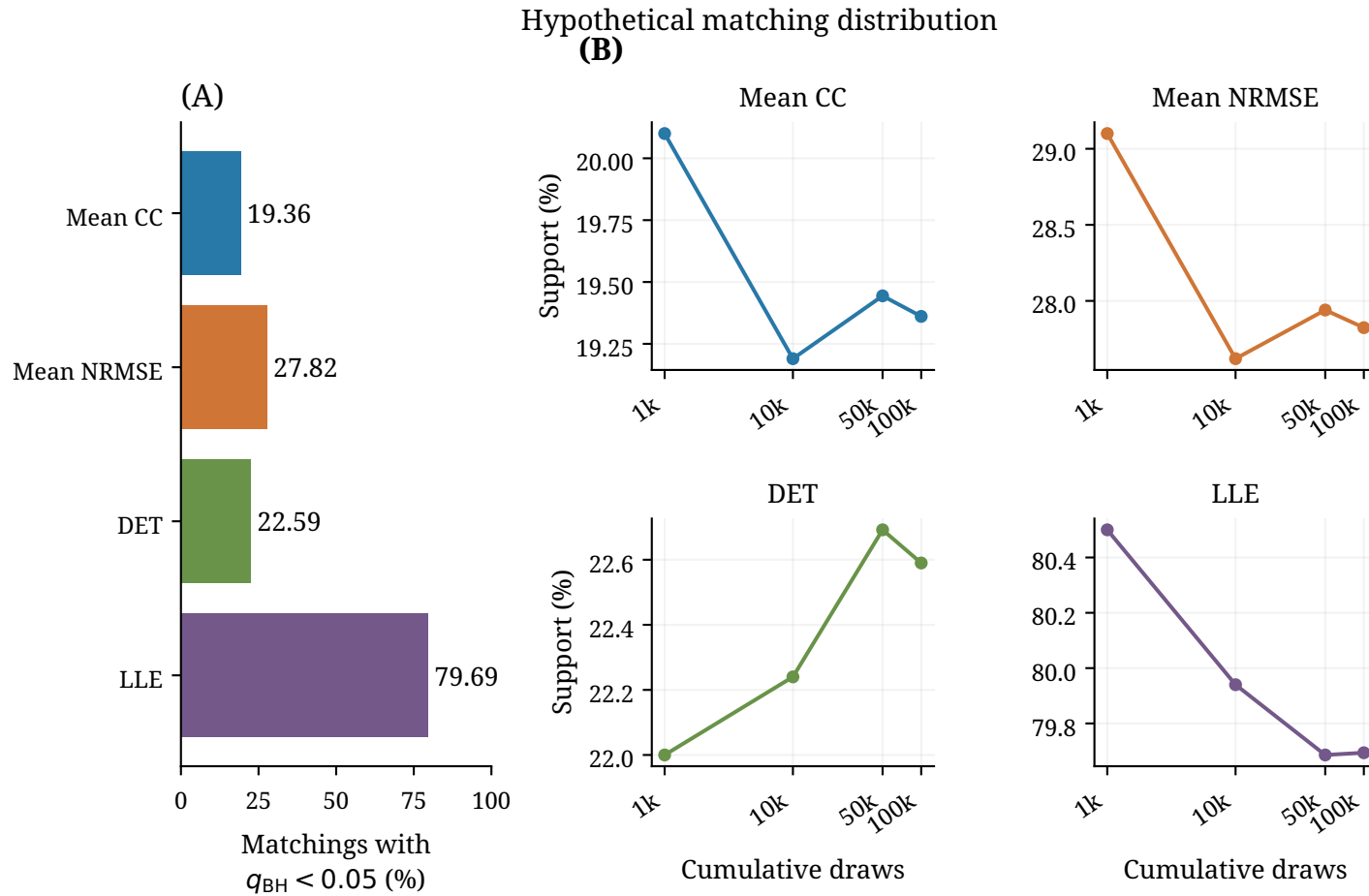


Figure S9: Hypothetical repeated-session dependence. (A) Fractions of 100,000 matchings with $q_{BH} < 0.05$ for the four headline metrics, applying BH7 within matching. All four median directions were preserved in every sampled matching. (B) Cumulative support frequencies at 1k/10k/50k/100k draws. These are summaries of a hypothetical matching distribution, not probabilities of true significance or participant recovery.

S14 Symbolic Dynamics

Exploratory symbolic analysis encoded PPG-derived peak-to-peak intervals (PPI) at 60/120/180 s. Peaks were detected without extra smoothing, using minimum separation $\lceil 0.30f_s \rceil$ samples and prominence 0.20 times the 95th–5th percentile amplitude range. PPI was the successive peak-index difference divided by f_s . Acceptance required finite PPI, no intervals outside 0.30–2.00 s, peak-span coverage $\geq 90\%$, and at least $\max\{3, \lceil T/2 \rceil - 2\}$ intervals for duration T seconds. Relative changes above 0.20 were diagnostic, not rejection criteria.

Within each accepted window, PPI was quantized into six equal-width levels between minimum and maximum. Internal-edge values entered the upper bin; constant sequences used level zero. Overlapping three-symbol words were assigned to 0V when both adjacent differences were zero, 1V when exactly one was zero, and 2V when both were nonzero. 2V combined same-sign 2LV and opposite-sign 2UV changes. Occurrence percentages summed to 100% within windows, although separately aggregated session-state medians need not. Paired comparisons followed session-state medians of accepted windows.

Table S13 retains all nine category–duration results, with BH3 within duration. Median 0V effects were positive and 2V effects negative at every duration; 1V was near zero and changed aggregate sign at 180 s. None passed its BH family. The 60-s 0V and 180-s 2V CIs excluded zero; the latter raw p was 0.048441 but q was 0.145323. These CI/raw-test features do not override the exploratory multiplicity result.

Paired effects kept the same sign across all three durations in 17/20 sessions for 0V, 11/20 for 1V, and 14/20 for 2V. Counts include consistently positive or negative sessions, regardless of the aggregate direction, and do not imply equal magnitude. Symbolic proportions provide directional context (Fig. S10) outside the primary four-metric claim; they are not direct sympathetic/vagal measures or validated ECG-derived autonomic measures.

Table S13: Exploratory symbolic supporting results.

Category	Duration (s)	Paired n	Δ (pp)	95% CI (pp)	Raw p	q_{BH}	r_{rb}	Direction count
0V	60	20	+4.04	[+0.35, +7.18]	0.053169	0.113777	+0.495	14/20
1V	60	20	+0.41	[-1.47, +2.23]	0.498009	0.498009	+0.181	10/20
2V	60	20	-0.91	[-6.40, +0.37]	0.075851	0.113777	-0.457	13/20
0V	120	20	+2.32	[-2.64, +5.63]	0.474905	0.712358	+0.190	13/20
1V	120	20	+0.03	[-1.67, +2.13]	0.927279	0.927279	+0.029	10/20
2V	120	20	-2.30	[-4.68, +1.78]	0.329983	0.712358	-0.257	12/20
0V	180	20	+4.58	[-1.12, +13.94]	0.164957	0.247436	+0.362	13/20
1V	180	20	-0.06	[-4.31, +1.95]	0.956329	0.956329	-0.019	10/20
2V	180	20	-2.67	[-5.37, -0.44]	0.048441	0.145323	-0.505	15/20

All nine tests are exploratory, with BH3 separately within duration. pp denotes percentage points. Direction counts are positive for 0V/1V and negative for 2V; 1V uses a positive reporting convention even at its slightly negative 180-s median.

Same-sign-across-duration counts are 17/20 (0V), 11/20 (1V), 14/20 (2V), including consistently opposite-direction sessions. Effects/CIs use two decimals, raw p/q six, and rank-biserial effects three.

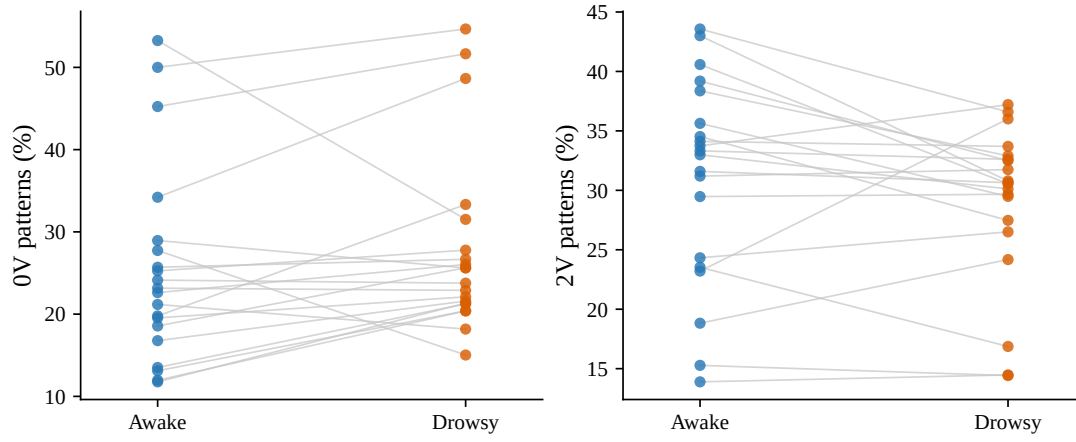
S15 Overall Robustness Summary

Table S14 separates nominal effects, direction preservation, uncertainty, and statistical support. Mean CC decreased and Mean NRMSE increased across tested window, label, delay, and dimension settings, with variable magnitudes/CIs and attenuated support under hypothetical grouping.

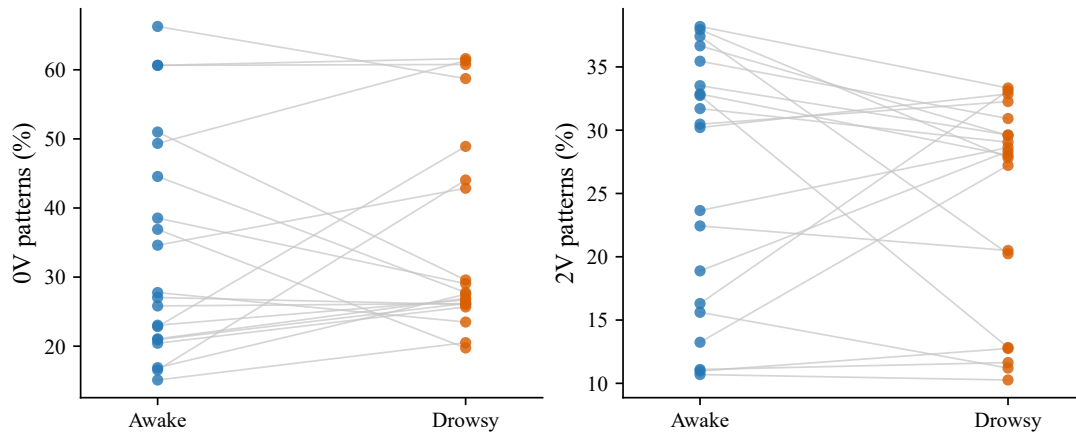
DET retained its nominal decrease across windows, labels, dimensions, and tested RQA RR/Theiler settings, but became near-zero and positive at non-nominal delays. Embedding-delay dependence is therefore an explicit qualification. LLE retained its decrease across all tested branches and the strongest support in hypothetical grouping, while fit/QC choices changed magnitude and accepted-window counts.

Symbolic evidence remains exploratory. Sensitivities are supportive analyses of the same recordings, rather than independent replications. Unknown participant–session linkage remains unresolved by hypothetical grouping. The matrix assigns no pooled significance, robustness score, or metric ranking.

(A) 60 s



(B) 120 s



(C) 180 s

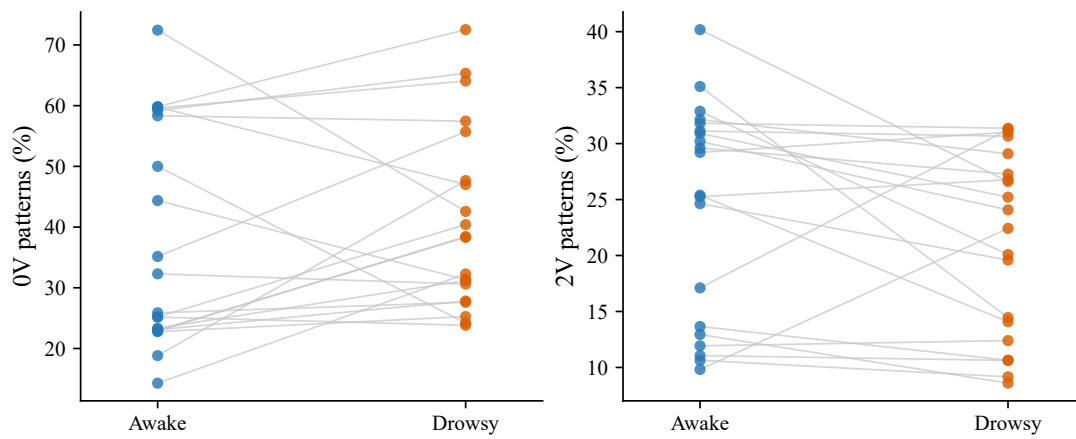


Figure S10: Exploratory symbolic PPI evidence at (A) 60, (B) 120, and (C) 180 s. Existing paired session-state 0V/2V occurrence proportions follow six-level PPI quantization and overlapping three-symbol words. Median 0V changes are positive and 2V changes negative, but none of the nine category-duration tests, including 1V in Table S13, passes within-duration BH3. Proportions provide supporting context, not direct sympathetic/vagal activity.

Table S14: Overall robustness evidence matrix.

Metric	Nominal 60-s	Window length	Label rules	τ/m	Metric-specific estimator settings	Hypothetical grouping	Interpretation limit
Mean CC	Decrease; BH support	Direction preserved; uncertainty varies	Direction preserved; T60 CI spans zero	Direction preserved; some CIs span zero	Not tested separately here	Direction preserved; support attenuated	Magnitude/support depend on setting; session-level forecastability
Mean NRMSE	Increase; BH support	Direction preserved; magnitude varies	Direction preserved; magnitudes differ	Direction preserved; some CIs span zero	Not tested separately here	Direction preserved; support attenuated	Magnitude/support depend on setting; session-level forecastability
DET	Decrease; BH support	Direction preserved; 30/180-s CIs span zero	Direction preserved; magnitudes differ	Delay-sensitive; dimension direction preserved	RR/Theiler direction preserved; RR=0.01 support weaker	Direction preserved; support attenuated	Delay-dependent recurrence-line organization; not physical determinism
LLE	Decrease; BH support	Direction preserved; uncertainty varies	Direction preserved; T30 CI spans zero	Direction preserved; magnitude varies	Direction preserved; fit/QC retention and magnitude vary	Direction preserved; strongest support retained in this analysis	Finite-window divergence; QC changes cohort; no participant inference
Symbolic	Exploratory; no BH support	0V/2V directional context at 60/120/180 s	Not tested	Not tested	Not tested	Not tested	Outside primary claim; no direct autonomic interpretation

Descriptors distinguish median direction, CI uncertainty, and correction-specific support within tested configurations. Untested cells do not imply stability. Symbolic results are outside the primary four-metric claim, and unknown participant linkage remains a limitation.